```
#Multi linear regression

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# data loading
dataset = pd.read_csv("insurance.csv")
dataset

#one-hot encoding
data=pd.get_dummies(dataset,drop_first=True)
data

#data split input and output
independent=data[['age', 'bmi', 'children', 'sex_male', 'smoker_yes','region_northwest','region_southeast','region_southwes
dependent=data[['expenses']]

#training and testing
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.10,random_state=0)

#Model creation with Grid Search
from sklearn.model_selection import GridSearchCV
from sklearn.linear_model import LinearRegression

# Define the hyperparameter grid
param_grid = {
    'fit_intercept': [True, False],
    'positive': [True, False]
}

# Initialize GridSearchCV with LinearRegression
grid_search = GridSearchCV(estimator=LinearRegression(), param_grid=param_grid, cv=5, scoring='r2')
grid_search.fit(x_train, y_train)

# Best estimator and predictions
regressor = grid_search.best_estimator_
y_pred = regressor.predict(x_test)

print("Best Parameters Found:", grid_search.best_params_)

#Evaluation metrics
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R2 Score:", r2)

#Prediction
predict=regressor.predict([[19,27.9,2,1,0,1,1,0]])
predict

#deployment
import pickle
filename="Insurance_expense.sav"
pickle.dump(regressor,open(filename,'wb'))

#Use in another file
loaded_model=pickle.load(open("Insurance_expense.sav",'rb'))
predict=loaded_model.predict([[19,27.9,2,1,0,1,1,0]])
predict

# Get values from the user

age = int(input("Enter your age: "))
bmi = float(input("Enter bmi value: "))
Child_count = int(input("Enter Children count: "))
sex = int(input("Enter Sex (0 or 1): "))
smoker_yes = int(input("Enter Smoker or not (0 or 1): "))
region_northwest = int(input("Enter northwest region (0 or 1): "))
region_southeast = int(input("Enter southeast region (0 or 1): "))
region_southwest = int(input("Enter southwest region (0 or 1): "))

# Pass the actual user-input variables into the model prediction
predict = loaded_model.predict([[age, bmi, Child_count, sex, smoker_yes, region_northwest, region_southeast, region_southwe
predict
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
Best Parameters Found: {'fit_intercept': True, 'positive': True}
```

```
R2 Score: 0.8226398084152773
Enter your age: 23
Enter bmi value: 34.5
Enter Children count: 2
Enter Sex (0 or 1): 1
Enter Smoker or not (0 or 1): 1
Enter northwest region (0 or 1): 1
Enter southeast region (0 or 1): 1
Enter southwest region (0 or 1): 1
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
array([[29549.88799937]])
```

```
#Decision tree

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# data loading
dataset = pd.read_csv("insurance.csv")
dataset

#one-hot encoding
data=pd.get_dummies(dataset,drop_first=True)
data

#data split input and output
independent=data[['age', 'bmi', 'children', 'sex_male', 'smoker_yes','region_northwest','region_southeast','region_southwes
dependent=data[['expenses']]

#training and testing
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.10,random_state=0)

#Model creation with Grid Search
from sklearn.model_selection import GridSearchCV
from sklearn.tree import DecisionTreeRegressor

# Define the hyperparameter grid for Decision Tree
param_grid = {
    'criterion': ['squared_error', 'friedman_mse', 'absolute_error', 'poisson'],
    'splitter': ['best', 'random'],
    'max_depth': [None, 5, 10, 15],
    'min_samples_split': [2, 5, 10]
}

# Initialize GridSearchCV with DecisionTreeRegressor
grid_search = GridSearchCV(estimator=DecisionTreeRegressor(random_state=0), param_grid=param_grid, cv=5, scoring='r2')
grid_search.fit(x_train, y_train)

# Best estimator and predictions
regressor = grid_search.best_estimator_
y_pred = regressor.predict(x_test)

print("Best Parameters Found:", grid_search.best_params_)

#Evaluation metrics
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R2 Score :", r2)

#Prediction
predict=regressor.predict([[19,27.9,2,1,0,1,1,0]])
predict

#deployment
import pickle
filename="Insurance_expense.sav"
pickle.dump(regressor,open(filename,'wb'))

#Use in another file
loaded_model=pickle.load(open("Insurance_expense.sav",'rb'))
predict=loaded_model.predict([[19,27.9,2,1,0,1,1,0]])
predict

# Get values from the user

age = int(input("Enter your age: "))
bmi = float(input("Enter bmi value: "))
Child_count = int(input("Enter Children count: "))
sex = int(input("Enter Sex (0 or 1): "))
```

```
smoker_yes = int(input("Enter Smoker or not (0 or 1): "))
region_northwest = int(input("Enter northwest region (0 or 1): "))
region_southeast = int(input("Enter southeast region (0 or 1): "))
region_southwest = int(input("Enter southwest region (0 or 1): "))

# Pass the actual user-input variables into the model prediction
predict = loaded_model.predict([[age, bmi, Child_count, sex, smoker_yes, region_northwest, region_southeast, region_southwe
predict
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
Best Parameters Found: {'criterion': 'absolute_error', 'max_depth': 5, 'min_samples_split': 2, 'splitter': 'best'}
R2 Score : 0.8910988958995868
Enter your age: 45
Enter bmi value: 23.5
Enter Children count: 3
Enter Sex (0 or 1): 1
Enter Smoker or not (0 or 1): 1
Enter northwest region (0 or 1): 1
Enter southeast region (0 or 1): 1
Enter southwest region (0 or 1): 1
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
array([35069.37])
```

```
#Random forest

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# data loading
dataset = pd.read_csv("insurance.csv")

#one-hot encoding
data=pd.get_dummies(dataset,drop_first=True)

#data split input and output
independent=data[['age', 'bmi', 'children', 'sex_male', 'smoker_yes','region_northwest','region_southeast','region_southwes
dependent=data[['expenses']]

#training and testing
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.10,random_state=0)

#Model creation with Grid Search
from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestRegressor

# Define the hyperparameter grid for Random Forest
param_grid = {
    'criterion': ['squared_error', 'friedman_mse', 'absolute_error', 'poisson'],
    'max_features': ['sqrt', 'log2', None],
    'n_estimators': [10, 50, 100]
}

# Initialize GridSearchCV with RandomForestRegressor
grid_search = GridSearchCV(estimator=RandomForestRegressor(random_state=0), param_grid=param_grid, cv=5, scoring='r2', n_jo
grid_search.fit(x_train, y_train)

# Best estimator and predictions
regressor = grid_search.best_estimator_
y_pred = regressor.predict(x_test)

print("Best Parameters Found:", grid_search.best_params_)

#Evaluation metrics
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R2 Score :", r2)

#Prediction
predict=regressor.predict([[19,27.9,2,1,0,1,1,0]])
predict

#deployment
import pickle
filename="Insurance_expense.sav"
pickle.dump(regressor,open(filename,'wb'))

#Use in another file
```

```
loaded_model=pickle.load(open("Insurance_expense.sav",'rb'))
predict=loaded_model.predict([[19,27.9,2,1,0,1,1,0]])
predict

# Get values from the user

age = int(input("Enter your age: "))
bmi = float(input("Enter bmi value: "))
Child_count = int(input("Enter Children count: "))
sex = int(input("Enter Sex (0 or 1): "))
smoker_yes = int(input("Enter Smoker or not (0 or 1): "))
region_northwest = int(input("Enter northwest region (0 or 1): "))
region_southeast = int(input("Enter southeast region (0 or 1): "))
region_southwest = int(input("Enter southwest region (0 or 1): "))

# Pass the actual user-input variables into the model prediction
predict = loaded_model.predict([[age, bmi, Child_count, sex, smoker_yes, region_northwest, region_southeast, region_southwe
predict
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/base.py:1389: DataConversionWarning: A column-vector y was passed when a 1d
  return fit_method(estimator, *args, **kwargs)
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
Best Parameters Found: {'criterion': 'absolute_error', 'max_features': 'log2', 'n_estimators': 100}
R2 Score : 0.9212641108968138
Enter your age: 34
Enter bmi value: 24.5
Enter Children count: 2
Enter Sex (0 or 1): 1
Enter Smoker or not (0 or 1): 1
Enter northwest region (0 or 1): 1
Enter southeast region (0 or 1): 1
Enter southwest region (0 or 1): 1
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names,
  warnings.warn(
array([20427.5534])
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

#data loading
dataset=pd.read_csv("insurance.csv")
#dataset

#one-hot encoding
dataset=pd.get_dummies(dataset,drop_first=True)
#dataset

#data split input and output
independent=datasetindependent=data[['age', 'bmi', 'children', 'sex_male', 'smoker_yes','region_northwest','region_southeas
dependent=data[['expenses']]

#training and testing
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.10,random_state=0)

#Model creation
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)


# Model creation with Grid Search
from sklearn.model_selection import GridSearchCV
from sklearn.svm import SVR



# Initialize GridSearchCV with SupportVectorRegressor
param_grid = {'kernel':['rbf','poly','sigmoid','linear'], 'C':[10,100,1000,2000,3000],'gamma':['auto','scale']}
grid_search.fit(x_train, y_train)

# Best estimator and predictions
regressor = grid_search.best_estimator_
y_pred = regressor.predict(x_test)

print("Best Parameters Found:", grid_search.best_params_)
```

```
#Evaluation metrics
from sklearn.metrics import r2_score
r2 = r2_score(y_test, y_pred)
print("R2 Score :", r2)

#Prediction
predict=regressor.predict([[19,27.9,2,1,0,1,1,0]])
predict

#deployment
import pickle
filename="Insurance_expense.sav"
pickle.dump(regressor,open(filename,'wb'))

#Use in another file
loaded_model=pickle.load(open("Insurance_expense.sav",'rb'))
predict=loaded_model.predict([[19,27.9,2,1,0,1,1,0]])
predict

# Get values from the user

age = int(input("Enter your age: "))
bmi = float(input("Enter bmi value: "))
Child_count = int(input("Enter Children count: "))
sex = int(input("Enter Sex (0 or 1): "))
smoker_yes = int(input("Enter Smoker or not (0 or 1): "))
region_northwest = int(input("Enter northwest region (0 or 1): "))
region_southeast = int(input("Enter southeast region (0 or 1): "))
region_southwest = int(input("Enter southwest region (0 or 1): "))

# Pass the actual user-input variables into the model prediction
predict = loaded_model.predict([[age, bmi, Child_count, sex, smoker_yes, region_northwest, region_southeast, region_southwe
predict
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:1408: DataConversionWarning: A column-vector y was passe
  y = column_or_1d(y, warn=True)
Best Parameters Found: {'C': 3000, 'gamma': 'scale', 'kernel': 'poly'}
R2 Score : 0.8874523481068479
Enter your age: 34
Enter bmi value: 34.5
Enter Children count: 3
Enter Sex (0 or 1): 1
Enter Smoker or not (0 or 1): 1
Enter northwest region (0 or 1): 1
Enter southeast region (0 or 1): 1
Enter southwest region (0 or 1): 1
array([4820749.2187167])
```

```
#Overall r_2 score value:
#Multiple linear regression  -  0.8226398084152773
#Decision tree               -  0.8910988958995868
#Random forest               -  0.9212641108968138
#SVR(poly)                   -  0.8874523481068479
```

```
pip install streamlit
```

```
Collecting streamlit
  Downloading streamlit-1.66.0-py3-none-any.whl.metadata (10 kB)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (8.5.0)
Requirement already satisfied: numpy<3,>=1.25.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dist-packages (from streamlit) (26.3)
Requirement already satisfied: pandas<4,>=1.5.3 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.3)
Requirement already satisfied: pillow<13,>=9.2.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (11.3.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.3-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.33.6)
Requirement already satisfied: pyarrow!=25.0.0,<27,>=10.0.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (23.
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.1
Requirement already satisfied: starlette<2,>=0.46.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (1.7.0)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.54.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.15.1)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.0.
Requirement already satisfied: websockets<18,>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->str
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5,>=4.0.0->streamlit) (3.20)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->str
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit) (2
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit)
```

```
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.13/dist-packages (from uvicorn<1,>=0.30.0->streamlit) (0.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->altair!=5.4.0,!=5.4.
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonsch
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=
Downloading streamlit-1.66.0-py3-none-any.whl (10.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.4/10.4 MB 75.6 MB/s eta 0:00:00
Downloading pydeck-0.9.3-py2.py3-none-any.whl (11.4 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 11.4/11.4 MB 90.9 MB/s eta 0:00:00
Installing collected packages: pydeck, streamlit
Successfully installed pydeck-0.9.3 streamlit-1.66.0
```

```
pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.13/dist-packages (2.1.3)
```

```
!pip install streamlit
```

```
Requirement already satisfied: streamlit in /usr/local/lib/python3.13/dist-packages (1.66.0)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (8.5.0)
Requirement already satisfied: numpy<3,>=1.25.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dist-packages (from streamlit) (26.3)
Requirement already satisfied: pandas<4,>=1.5.3 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.3)
Requirement already satisfied: pillow<13,>=9.2.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (11.3.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.9.3)
Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.33.6)
Requirement already satisfied: pyarrow!=25.0.0,<27,>=10.0.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (23.
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.1
Requirement already satisfied: starlette<2,>=0.46.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (1.7.0)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.54.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.15.1)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.0.
Requirement already satisfied: websockets<18,>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->str
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5,>=4.0.0->streamlit) (3.20)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->str
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit) (2
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.13/dist-packages (from uvicorn<1,>=0.30.0->streamlit) (0.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->altair!=5.4.0,!=5.4.
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonsch
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=
```

```
pip install pyngrok
```

```
Collecting pyngrok
  Downloading pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.13/dist-packages (from pyngrok) (6.0.3)
Downloading pyngrok-8.1.2-py3-none-any.whl (25 kB)
Installing collected packages: pyngrok
Successfully installed pyngrok-8.1.2
```

```
%%writefile app.py
import streamlit as st
import pandas as pd
import pickle

# Load trained model
loaded_model = pickle.load(open("Insurance_expense.sav", "rb"))

# Page title
st.title("💾 Insurance Expense Prediction")
st.write("Enter the customer details to predict insurance expenses.")

# User inputs
age = st.number_input(
    "Age",
```

```
    min_value=1,
    max_value=100,
    value=25
)

bmi = st.number_input(
    "BMI",
    min_value=10.0,
    max_value=60.0,
    value=25.0
)

children = st.number_input(
    "Number of Children",
    min_value=0,
    max_value=10,
    value=0
)

# Sex
sex = st.selectbox(
    "Sex",
    ["Female", "Male"]
)

# Smoker
smoker = st.selectbox(
    "Smoker",
    ["No", "Yes"]
)

# Region
region = st.selectbox(
    "Region",
    ["Northeast", "Northwest", "Southeast", "Southwest"]
)

# Convert categorical values to 0/1
sex_male = 1 if sex == "Male" else 0

smoker_yes = 1 if smoker == "Yes" else 0

region_northwest = 1 if region == "Northwest" else 0
region_southeast = 1 if region == "Southeast" else 0
region_southwest = 1 if region == "Southwest" else 0

# Prediction button
if st.button("🔮 Predict Expenses"):

    # Create DataFrame with SAME column names used during training
    new_data = pd.DataFrame({
        'age': [age],
        'bmi': [bmi],
        'children': [children],
        'sex_male': [sex_male],
        'smoker_yes': [smoker_yes],
        'region_northwest': [region_northwest],
        'region_southeast': [region_southeast],
        'region_southwest': [region_southwest]
    })

    # Prediction
    prediction = loaded_model.predict(new_data)

    # Display result
    st.success(
       f"💰 Predicted Insurance Expense: ${float(prediction.item()):,.2f}"
    )
```

```
Writing app.py
```

```
!pip install streamlit pyngrok
```

```
Requirement already satisfied: streamlit in /usr/local/lib/python3.13/dist-packages (1.66.0)
Requirement already satisfied: pyngrok in /usr/local/lib/python3.13/dist-packages (8.1.2)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (8.5.0)
Requirement already satisfied: numpy<3,>=1.25.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dist-packages (from streamlit) (26.3)
Requirement already satisfied: pandas<4,>=1.5.3 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.3)
Requirement already satisfied: pillow<13,>=9.2.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (11.3.0)
Requirement already satisfied: pydeck<1,>=0.8.0b4 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.9.3)
Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.33.6)
```

```
Requirement already satisfied: pyarrow!=25.0.0,<27,>=10.0.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (23.
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.1
Requirement already satisfied: starlette<2,>=0.46.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (1.7.0)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.54.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.15.1)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.0.
Requirement already satisfied: websockets<18,>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.13/dist-packages (from pyngrok) (6.0.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5.0.0->str
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=5
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!=5.4.1,<7,>=
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5,>=4.0.0->streamlit) (3.20)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->str
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit) (2
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.5.3->streamlit)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->stream
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.13/dist-packages (from uvicorn<1,>=0.30.0->streamlit) (0.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->altair!=5.4.0,!=5.4.
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonsch
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas<4,>=
```

```
!streamlit run app.py ^&>/content/logs.txt&
```

```
from pyngrok import ngrok
ngrok.kill()
```

```
from pyngrok import ngrok
ngrok.set_auth_token("your token number")
public_url=ngrok.connect(8501)
print(public_url)
```

```
NgrokTunnel: "https://herself-lapping-kindle.ngrok-free.dev" -> "http://localhost:8501"
```